

Ex No: 1

IMPLEMENT EIGHT QUEENS PROBLEM

Aim:

To Implement Eight Queens Problem using backtracking

Procedure:

Program

N = 8 # Size of the chessboard (8x8)

Function to print the solution

def printSolution(board):

for row in board:

for i in range(N):

print("Q" if row[i] else ".", end=" ")

print()

print() # Add a newline for readability

Function to check if a queen can be placed on board[row][col]

def isSafe(board, row, col):

Check the column

for i in range(row):

if board[i][col]:

return False

```

# Check the upper left diagonal
for i, j in zip(range(row - 1, -1, -1), range(col - 1, -1, -1)):
    if board[i][j]:
        return False

# Check the upper right diagonal
for i, j in zip(range(row - 1, -1, -1), range(col + 1, N)):
    if board[i][j]:
        return False
return True

# Function to solve the 8 Queens problem using backtracking
def solve(board, row):
    if row == N:
        printSolution(board)
        return True # Stop after finding one solution

    for col in range(N):
        if isSafe(board, row, col):
            board[row][col] = 1 # Place queen

            if solve(board, row + 1): # Recur to place next queen
                return True # Stop after first valid solution

            board[row][col] = 0 # Backtrack (remove queen)

    return False

# Main function to initialize the board and start solving the problem
def eightQueens():
    board = [[0 for _ in range(N)] for _ in range(N)]

```

```
if solve(board, 0):  
    print("One possible solution is:")  
else:  
    print("Solution does not exist")  
  
# Calling the function  
eightQueens()
```

Output:

One of the possible solutions is:

```
. . . . Q . . .  
. . . Q . . . .  
. . . . Q . . .  
Q . . . . . . .  
. . . . . Q .  
. Q . . . . . .  
. . Q . . . . .  
. . . . . . Q
```

Total valid solutions found: 92

Example Solution Matrix:

Q	.	.	.	.	.	.	.
.	.	.	.	Q	.	.	.
.	.	.	.	.	.	.	Q
.	.	.	.	.	Q	.	.
.	.	Q	.	.	.	.	.
.	.	.	.	.	.	Q	.
.	Q	.	.	.	.	.	.
.	.	.	Q	.	.	.	.

=== Code Execution Successful ===